

Vincent Wang

Robotics Systems | VLA-WAM

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EDUCATION

University of Pennsylvania

Expected May 2027

M.S. in Mechanical Engineering and Applied Mechanics - Mechatronics and Robotics | GPA: 3.81/4.0

- Coursework: Advanced Dynamics, Mechatronic Systems, Engineering Mathematics, Operating Systems.

East China University of Science and Technology

June 2025

B.Eng. in Intelligent Manufacturing Engineering | GPA: 3.6/4.0 | Top 15%

EXPERIENCE

Synthoid.ai | VLA Research Intern

May 18 - Aug 21, 2026

Real-robot infrastructure, VLA training, and independent policy-distillation research

- Independently built SDK control and a VLA adapter for a 7-DoF industrial arm, adding Cartesian control, multi-rate interpolation, and automatic recovery; delivered stable teleoperation.
- Led dexterous-hand teleoperation and LeRobot data collection with head/wrist cameras; synchronized timestamps and screened trajectories using first- through third-order action derivatives.
- Resolved abnormal motion from 7-DoF IK ambiguity and traced replay errors to an incorrect URDF.
- Deeply contributed to proprietary VLA pre-training through implementation/configuration, distributed runs, checkpoint management, ablations, and debugging; found a gripper output-contract mismatch and GPU ECC failures.
- Independently led Path-OPD from problem formulation and method design through implementation, matched experiments, and manuscript writing.

Wuji Tech | Robotics Control Intern

July - Sept 2024

Dexterous-hand teleoperation and low-level motor control

- Re-architected an unstable impedance loop into a torque-feedback admittance controller in MATLAB/Simulink for dexterous-hand teleoperation.
- Built dexterous-hand connector components and Linux C++ host software for reliable multi-motor CAN control.

SELECTED PROJECTS & LEADERSHIP

ECUST Robocon Team | Co-founder & Control Lead

Sept 2022 - July 2024

Autonomous robot systems and interdisciplinary technical leadership

- Built the team's first STM32 motion-control system and integrated Point-LIO, embedded YOLO (approximately 20 FPS), MoveIt, and chassis control into an autonomous robot stack.
- Delivered fixed-scene SLAM navigation, target recognition, and object grasping.
- Led a 20-member interdisciplinary team, owned system architecture and technology selection, and trained 60 new members.

MEAM 510 | Autonomous Mobile Robot

2025

Embedded autonomous robot system

- Built a closed-loop ESP32/FreeRTOS mobile robot with sensing, Vive/EKF localization, A* planning, chassis control, mission FSM, and system validation.

CIS 5480 | PennOS

2026

Operating-systems engineering

- Implemented and debugged PennOS in C, including processes, scheduling, system calls, signals, a file system, and a shell.

RESEARCH & PUBLICATIONS

Path-OPD - Online Policy Distillation for Flow-Matching VLAs

Manuscript in preparation

- First author; independently led formulation, design, implementation, matched experiments, and writing; targeting ICLR 2027.

MDATS - Multi-Domain Adaptive Classification for Mechanical Fault Detection

IEEE MEAE 2024 • Published

- First author; achieved 88.27% target-domain accuracy, 40.6 points over baseline, with a PyTorch pipeline.

TECHNICAL SKILLS

Programming & Systems: Python, C/C++, Linux, Git, ROS 2, CAN

Robotics & Control: Teleoperation, impedance/admittance control, STM32/ESP32, FreeRTOS, MoveIt

VLA & Training: PyTorch, distributed training, VLA post-training, flow matching, online policy distillation, WAM

Robot Data & Evaluation: LeRobot, multi-camera data, timestamp synchronization, trajectory screening, closed-loop evaluation

Language: GRE 324, fluent English communication

Vibe Coding: Symphony, harness